

ELEC 1100 Laboratory 2: Summary Sheet

Name: _____

Student ID: _____

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Notes:

- (1) Show demo to your TA for an attendance record; **bonus mark** will be given to **groups** who can complete all TA Checks in the first 2 hours of your enrolled lab session.

LA2 (Fri, before 15:30), LA3 (Mon, before 12:30), LA1 (Tue, before 15:30)

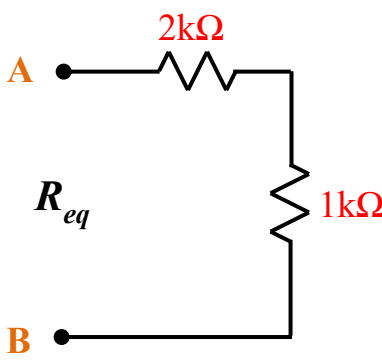
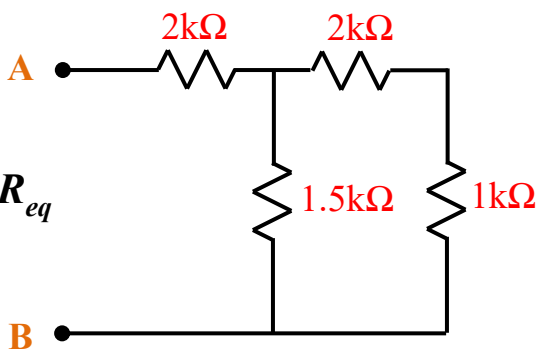
- (2) **Individually** submit your completed lab summary sheet to Canvas (LA1/LA2/LA3/LA4) "Assignments" before the deadline.

LA2 (Fri, deadline 16:20), LA3 (Mon, deadline 13:20), LA1 (Tue, deadline 16:20)

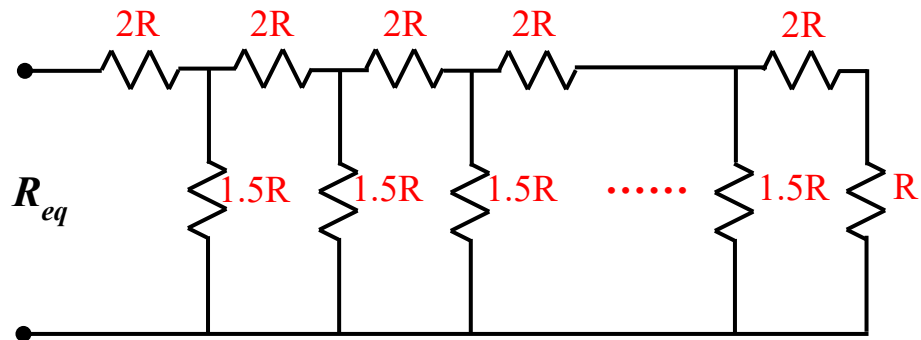
Experimental Results and Analysis

Q1: Calculate for R_{eq} and complete the table below.

Q2: Record the value of R_{eq} displayed on the screen (including the unit $\Omega/k\Omega$) to fill in the table below.

Figure 1	Figure 2
	
By calculation: $R_{eq} =$ (show your calculation steps)	By calculation: $R_{eq} =$ (show your calculation steps)
From the multimeter: $R_{eq} =$ This value should match your calculation above.	From the multimeter: $R_{eq} =$ This value should match your calculation above.

Q3: Based on what you got above, analysis a resistor network with infinite modules (R , $1.5R$, $2R$) as below. Prove that $R_{eq} = 3R$. Show clearly the calculation steps.



Q4: Record the measurement of voltages V_{AD} , V_{AB} , V_{BC} , V_{CD} and V_{DA} to fill in the table below.

Q5: Complete the table to show that the sum of the voltages ACROSS all of the components (including the power source) in a loop complies with Kirchhoff's voltage law (KVL).

V_{AD}	V_{AB}	V_{BC}	V_{CD}	V_{DA}
$V_{AD} = V_{AB} + V_{BC} + V_{CD}$ (use the measured voltages to verify the equation) $V_{AD} =$ $V_{AB} + V_{BC} + V_{CD} =$				
$V_{AB} + V_{BC} + V_{CD} + V_{DA} = 0$ (use the measured voltages to verify the equation) $V_{AB} + V_{BC} + V_{CD} + V_{DA} =$				

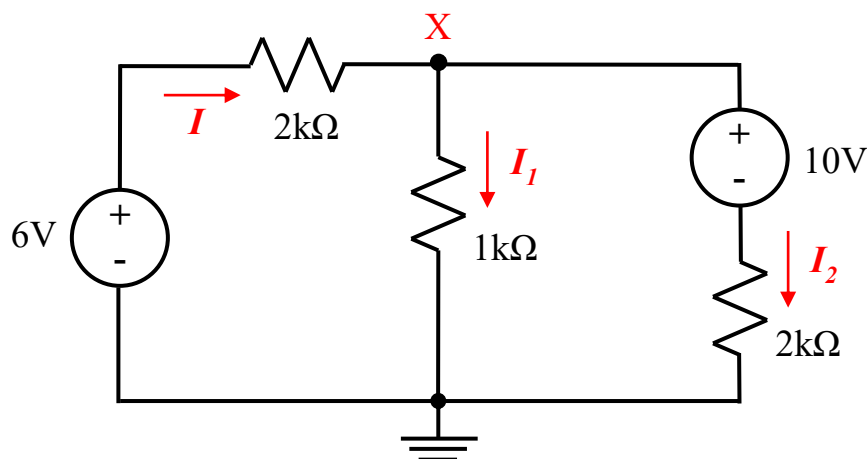
Q6: Calculate for branch currents I_1 , I_2 and I_3 to fill in the table below.

Q7: Record the value of I displayed on the screen (including the unit “mA”). Complete the table to show that the sum of currents complies with Kirchhoff’s current law (**KCL**).

$$I = I_1 + I_2 + I_3$$

By calculation: $I_1 =$ _____ $I_2 =$ _____ $I_3 =$ _____ $I_1 + I_2 + I_3 =$ _____
From the multimeter: Total current $I =$ _____ This value should match your calculation above.

Q8: Use **KCL** to solve the currents I , I_1 and I_2 , then determine the voltage at node X in the circuit below.



Show your calculation steps. **Notice the unit and direction of each value.

$I =$ _____ $I_1 =$ _____ $I_2 =$ _____ $V_X =$ _____

Q9: Record the value of V_X from the multimeter to fill in the table below.

Q10: Record the value of I from the multimeter (including the unit “mA”) to fill in the table below.

From the multimeter: $V_X =$ This value should match your calculated V_X in Q8.	From the multimeter: $I =$ This value should match your calculated I in Q8.
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Q11: Record the value of R_{eq} from the multimeter (including the unit $\Omega/k\Omega$) to fill in the table.

From the multimeter: $R_{eq} =$
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Q12: Record the voltages from the multimeter and fill in the table.

From the multimeter: $V_C =$ (versus GND)	From the multimeter: $V_D =$ (versus GND)
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Q13: Record the value of I from the multimeter (including the unit “mA”) to fill in the table.

From the multimeter: $I =$

Q14: Based on your record in Q13, use **Ohm’s law** to calculate for R_{eq} to fill in the table.

By calculation (including the unit $\Omega/k\Omega$): $R_{eq} = \frac{V}{I} =$ This value should match your measured R_{eq} in Q11.
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